Web Development using PHP

Client-side Scripting – Javascript

Introduction to Object Oriented Programming

(Lecture 5)

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1. Introducing Javascript

- JavaScript was designed to add interactivity to HTML pages.
- JavaScript is a scripting language.
- JavaScript is usually embedded directly into HTML pages.
- JavaScript is an interpreted language (means that scripts execute without preliminary compilation).
- It is an event-driven language.

To insert a JavaScript into an HTML page, use the <script> tag.

Inside the <script> tag use the type attribute to define the scripting language.

The <script> and </script> tells where the JavaScript starts and ends:

```
<html>
<body>
<h1>My First Web Page</h1>

<p id="demo">This is a paragraph.</p>

<script type="text/javascript">
... some JavaScript code ...
</script>

</body>
</html>
```

The lines between the <script> and </script> contain the JavaScript and are executed by the browser.

2. How to use javascript

The common way to use the javascript is as follows;

1. Define some HTML tag with ID attribute value given.
2. In that HTML tag add/call the function statement and bind it to some event on which you want to invoke the javascript function (event handler).
3. Write the said function (event handler) between <script> and </script> tags.
4. When the page downloads, your handler will be called on event you described within the HTML tag.

Let's practice these steps in the following examples;

2.1 Step 1 – define ID attribute in an HTML tag.

Create a page with example2.1.php in your htdocs/training/lecture5 directory and type the following code;

Example 2.1

```
<html>
<body>
<h1>My First Javascript Example</h1>
<br><br>

<input type="button" name="js_btn" id="js_btn" value="Click Me to get a Message!"/>

</body>
</html>
```

Run the page in the browser a button having label "Click Me to get a Message!" should display.

2.2 Step 2 – add/call the function statement in HTML tag.

Save example2.1.php as example2.2.php in your htdocs/training/lecture5 directory and type the (bold) line as shows follows;

Example 2.2

```
<html>
<body>
<h1>My First Javascript Example</h1>
<br><br>

<input type="button" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick="show_msg();" />

</body>
</html>
```

2.3 Step 3 – writing event handler in <script> tag.

Save example2.2.php as example2.3.php in your htdocs/training/lecture5 directory and type the (bold) line as shows follows;

Example 2.3

```
<html>
<script type="text/javascript">
function show_msg()
{
        alert("Have a nice day!!");
}
</script>
<body>
<h1>My First Javascript Example</h1>
<br><br>

<input type="button" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick="show_msg();" />

</body>
```

```
</html>
```

Run the page in the browser and click on the button an alert box should pop-up to display a message. This way you can use javascript in your web page.

Let's extend our example and display the text written by a user in a text field as follows;

Example 2.4

```
<html>
<script type="text/javascript">
function show_msg()
{
        alert("Have a nice day!!");
}
</script>
<body>
<h1>My First Javascript Example</h1>
<br>
write your comments: <input type="text" name="comments" id="comments" size="50">
<br><br>
<input type="button" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick="show_msg();" />

</body>
</html>
```

Run the page in the browser. This should display a text field in which you can provide any comments.

2.4 Introducing document 'getElementById()' method

The getElementById() method accesses the first element with the specified id.

Syntax

document.getElementById("*id*");

Parameter	**Description**
id	Required. The id of the element you want to access/manipulate

Let's extend our function and use getElementById() so it could capture and display the comments provided by user in the text field as follows;

Example 2.5

```
<html>
<script type="text/javascript">
function show_msg()
{
        var cmnts = document.getElementById('comments').value;
        alert(cmnts);
}
</script>
<body>
```

```
<h1>My First Javascript Example</h1>
<br>
write your comments: <input type="text" name="comments" id="comments" size="50">
<br><br>
<input type="button" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick="show_msg();" />

</body>
</html>
```

Run the page in the browser. After providing some text in comments field and upon clicking the button, it should display those comments in alert box.

Now let's add more functionality to our function for emptying the comments text field value after the alert has been appeared.

Example 2.6

```
<html>
<script type="text/javascript">
function show_msg()
{
        var cmnts = document.getElementById('comments').value;
        alert(cmnts);
        document.getElementById('comments').value = ' ';
}
</script>
<body>
<h1>My First Javascript Example</h1>
<br>
write your comments: <input type="text" name="comments" id="comments" size="50">
<br><br>
<input type="button" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick="show_msg();" />

</body>
</html>
```

Run the page in the browser. After providing some text in comments field and upon clicking the button, it should display those comments in alert box and when you press OK in the alert box, the value provided in comments field should disappear.

2.5 Validation through Javascript

Javascript can be used to validate user input at client-side i.e. before sending the page to server. In the following example, we'll create a form and display the comments to next page using POST method.

Save example 2.6 as example2.7.php and type the following bold lines;

Example 2.7

```
<html>
<script type="text/javascript">
function show_msg()
```

```
{
        var cmnts = document.getElementById('comments').value;
        alert(cmnts);
        //document.getElementById('comments').value = ' ';
}
</script>
<body>
<form action="user_reply.php" method="POST">
<h1>My First Javascript Example</h1>
<br>
write your comments: <input type="text" name="comments" id="comments" size="50">
<br><br>
<input type="submit" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick="show_msg();" />
</form>
</body>
</html>
```

Now create user_reply.php and display the comments provided by user there.

Now if the user does not provide any comments and presses the button, the form has been submitted without checking that user has entered some comments or not. Fortunately with javascript we can restrict the form's submission until the comments has not been provided. How to do that, let's see in the following example2.8.php;

Example 2.8

```
<html>
<script type="text/javascript">
function show_msg()
{
        var cmnts = document.getElementById('comments').value;
        //alert(cmnts);
        if(cmnts == ' ')
        {
                alert('Please provide some comments. Thank you!!');
                return false;
        }
        return true;
}
</script>
<body>
<form action="user_reply.php" method="POST">
<h1>My First Javascript Example</h1>
<br>
write your comments: <input type="text" name="comments" id="comments" size="50">
<br><br>
<input type="submit" name="js_btn" id="js_btn" value="Click Me to get a Message!" onclick=" return show_msg();" />
</form>
</body>
</html>
```

Run the page in the browser and click the button without providing any comments, you will see that the form has not been submitted and a help message is displayed to the user. After you provided the comments, no message displayed and the form takes you to the next page successfully.

Okay up till now, we've learned the novice programming for server-side and client-side scripting using PHP and Javascript respectively. So hopefully you are now able to develop some dynamic pages using PHP and javascript. Now I would like to explain concept of object oriented programming technique which actually reduces your effort in writing and managing the same code again and again.

3. Object Oriented Programming (OOP)

Object oriented programming is a style of coding in which related actions are grouped into classes to write and manage compact and effective code. It is actually a way of coding that allows developers to group similar tasks into **classes**. The basic aim behind this technique is

1. re-usability with minimum effort
2. easy-to-maintain piece of code

The major benefit gained over the structured programming technique is that, if a piece of information changes in your program, usually **only one change is required to update the code.**

3.1 Understanding Objects and Classes

Usually most people think Objects and Classes are the same thing or they mixed the concept by saying both are refer to same concept. However, this is not the case and there lies a slight difference between these two as one is (Class) is actually refer to ***sketching the code*** and the other (Objects) refers to actual ***usage (or bring in action) the code which has sketched in a Class.*** This can be explained by a real life example as follows;

A class, for example, is like **a Map or a blueprint of a house**. It defines the shape of the house on paper, with relationships between the different parts of the house clearly defined and planned out, even though the house doesn't exist.

An object, then, is like **the actual house** built according to that *map or blueprint*. The data stored in the object is like the wood, wires, and concrete which compose the house: without being assembled according to the blueprint, when it all comes together, it becomes an organized, useful house.

So **Classes form or build the structure of data and describe the ways to use that information/data to build objects.** More than one object can be built from the same class at the same time (*just like more than one house is created with single map, each with the same look but independent of the others having different families and decorations inside).*

3.2 Creating a Class

The syntax to create a class is simple: declare a class using the `class` keyword, followed by the name of the class and a set of curly braces (`{}`).

Example 3.1

```
<html>
<body>
<?php
class  my_lectures
{
   // Class properties and methods go here
}

?>
</body>
</html>
```

After creating the class, a new class can be instantiated and stored in a variable using the new keyword.

Example 3.2

```
<html>
<body>
<?php
class my_lectures
{
   // Class properties and methods go here
}

$obj = new my_lectures;
?>
</body>
</html>
```

To see the contents of the class, use var_dump() as follows;

Example 3.3

```
<html>
<body>
<?php
class my_lectures
{
   // Class properties and methods go here
}

$obj = new my_lectures;
var_dump($obj);
?>
</body>
</html>
```

Run the example in your browser, you should see the following output;

```
object(my_lectures)#1 (0) { }
```

In its simplest form, we've just completed our first OOP script.

4. Resources

4.1 websites

http://www.w3schools.com/

4.2 Books

Beginning PHP5 by Wrox word press Ltd.